

SIMATS ENGINEERING

ASSESSMENT TOOL 1- Enumeration Analysis

ETHICAL-HACKING

ITA1408

QUESTION 1

A cyber security consulting company has been contracted to perform an initial security assessment of the publicly accessible server `scanme.nmap.org`. Before vulnerability assessment, the client requires a detailed enumeration report. Using Nmap, perform host discovery, TCP port scanning, service version detection, operating system fingerprinting, and default script scanning. Analyze identified ports/services, determine their business purpose, evaluate whether they are necessary for public access, and assess how the information could support further attacks.

Sol :

To perform an authorized Nmap-based security enumeration of the specified target, identify exposed ports and services, understand their purpose, assess the attack surface, and recommend controls to reduce unnecessary exposure.

Command :

```
nmap scanme.nmap.org
```

```
nmap -sV scanme.nmap.org
```

```
sudo nmap -O scanme.nmap.org
```

```
sudo nmap -A scanme.nmap.org
```

Observation :

- The host is reachable and responds to Nmap probing.
- HTTP (80) and SSH (22) are common exposed services in the scan; additional ports may appear depending on the current target configuration.
- Service/version detection provides information about software running behind the ports.
- OS detection and aggressive/default-script scanning can reveal useful platform and configuration information.

Risks :

- Publicly exposed services increase the attack surface.
- Service banners and OS information can help attackers identify likely weaknesses.
- SSH may be targeted through weak credentials or outdated software if poorly secured.
- A public web service may expose vulnerable applications, configurations or outdated components.
- Enumeration information can support reconnaissance and later unauthorized-access attempts.

Security Recommendations :

- Keep all exposed services patched and supported.
- Allow only ports required for the server's business function.
- Restrict SSH to trusted networks/VPN where possible and use strong authentication.
- Disable unused services and remove unnecessary software.
- Use firewalls, logging, monitoring and periodic vulnerability assessments.
- Review public exposure regularly and perform authorized penetration testing.

Report / Execution Photo :

Illustrative terminal photo for the above command. Replace this image with your own actual Nmap execution screenshot before submission.

```
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
kali@kali:~$ nmap -sn scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 06:56 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.00061s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Nmap done: 1 IP address (1 host up) scanned in 8.34 seconds

kali@kali:~$ nmap -ss scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 06:57 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0011s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 906 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
53/tcp    closed domain
80/tcp    open  http
139/tcp    closed netbios-ssn
143/tcp    closed imap
161/tcp    closed xmpp
443/tcp    closed https
580/tcp    closed isakmp
993/tcp    closed imaps
3389/tcp   closed ms-wbt-server
5060/tcp   open  sip
5061/tcp   open  sip-tls
8080/tcp   closed http-proxy
8081/tcp   closed blackice-icecap

Nmap done: 1 IP address (1 host up) scanned in 104.31 seconds

kali@kali:~$ nmap -sV scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 06:59 EDT
Stats: 0:00:05 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan
Ping Scan Timing: About 100.00% done; ETC: 00:59 (0:00:00 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0059s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
80/tcp    open  http        Apache httpd 2.4.7 ((Ubuntu))
139/tcp    closed netbios-ssn
143/tcp    closed imap
443/tcp    closed https
3389/tcp   closed ms-wbt-server
5060/tcp   open  sip?
5061/tcp   open  sip-tls?
8080/tcp   closed http-proxy

Nmap done: 1 IP address (1 host up) scanned in 98.98 seconds

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 98.98 seconds

kali@kali:~$ sudo nmap -O scanme.nmap.org
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 07:01 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0080s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 986 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
53/tcp    closed domain
80/tcp    open  http
139/tcp    closed netbios-ssn
143/tcp    closed imap
443/tcp    closed https
993/tcp    closed imaps
3389/tcp   closed squid-http
3389/tcp   closed ms-wbt-server
5060/tcp   open  sip
5061/tcp   open  sip-tls
8080/tcp   closed http-proxy
8081/tcp   closed blackice-icecap
8085/tcp   closed unknown
OS fingerprint not ideal because: maxTimingRatio (3.404000e+00) is greater than 1.4
No OS matches for host

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 80.24 seconds

kali@kali:~$ nmap -sC scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 07:03 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.018s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
5060/tcp   open  sip
5061/tcp   open  sip-tls

Nmap done: 1 IP address (1 host up) scanned in 23.26 seconds

kali@kali:~$
```

```
Session Actions Edit View Help
5061/tcp open  sip-tls?
8080/tcp closed http-proxy
8081/tcp closed blackice-icecap

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 98.98 seconds

kali@kali:~$ sudo nmap -O scanme.nmap.org
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 07:01 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0080s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 986 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
53/tcp    closed domain
80/tcp    open  http
139/tcp    closed netbios-ssn
143/tcp    closed imap
443/tcp    closed https
993/tcp    closed imaps
3389/tcp   closed squid-http
3389/tcp   closed ms-wbt-server
5060/tcp   open  sip
5061/tcp   open  sip-tls
8080/tcp   closed http-proxy
8081/tcp   closed blackice-icecap
8085/tcp   closed unknown
OS fingerprint not ideal because: maxTimingRatio (3.404000e+00) is greater than 1.4
No OS matches for host

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 80.24 seconds

kali@kali:~$ nmap -sC scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 07:03 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.018s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
5060/tcp   open  sip
5061/tcp   open  sip-tls

Nmap done: 1 IP address (1 host up) scanned in 23.26 seconds

kali@kali:~$
```

Conclusion :

Nmap enumeration provides a clear view of the server's public attack surface. Reducing unnecessary services, patching exposed software, restricting administration and monitoring activity can significantly reduce risk.

QUESTION 2

The security team has deployed a deliberately vulnerable Linux server, 192.168.56.101 (Metasploitable 2), in an isolated laboratory. As part of pre-engagement reconnaissance, use Nmap to perform comprehensive enumeration. Identify all open ports, running services, service versions and operating system information. Analyze each service, categorize it into High, Medium and Low risk based on exposure and functionality, and explain how enumeration guides a penetration tester in selecting potential attack vectors and exploitation strategies.

Sol :

To perform an authorized Nmap-based security enumeration of the specified target, identify exposed ports and services, understand their purpose, assess the attack surface, and recommend controls to reduce unnecessary exposure.

Command :

```
sudo nmap -sS -sV -O -p- 192.168.56.101
```

```
sudo nmap -sC -sV 192.168.56.101
```

Observation :

- The deliberately vulnerable VM can expose many services at the same time.
- Typical lab findings may include FTP, SSH, Telnet, SMTP, HTTP, SMB/NetBIOS and MySQL.
- Service/version enumeration helps determine what software is installed and whether legacy services are present.
- The exact ports and versions must be taken from the student's actual VM scan.

Risks :

- **High:** legacy/insecure remote services, exposed databases and broadly accessible file-sharing services.
- **Medium:** required services such as SSH or HTTP that are reachable but properly restricted and authenticated.
- **Low:** necessary services that are patched, tightly filtered and monitored.
- Multiple exposed services provide more possible entry points and increase attack surface.

Security Recommendations :

- Keep Metasploitable isolated from production and the public Internet.
- In real systems, disable legacy services such as Telnet and unnecessary FTP.
- Restrict database and file-sharing services to trusted hosts.
- Use firewalls and network segmentation.
- Patch production systems and remove known-vulnerable software.
- Use enumeration results to prioritize defensive remediation and authorized testing.

Report / Execution Photo :

Illustrative terminal photo for the above command. Replace this image with your own actual Nmap execution screenshot before submission.

```
Firefox ESR
Browse the World Wide Web

[kali@kali:~]$ ping -c 4 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data:
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=1.79 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=1.73 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=1.62 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=2.31 ms

--- 192.168.56.102 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 1.72/2.716/3.193/0.773 ms

[kali@kali:~]$
[kali@kali:~]$ sudo nmap -iR 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2020-09-18 08:28 EDT
Nmap scan report for 192.168.56.102
Host is up (0.0017s latency).
NAC Address: 00:00:27:71:08:10 (PCN Systemtechnika/Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 8.16 seconds

[kali@kali:~]$
[kali@kali:~]$ sudo nmap -iR -p- 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2020-09-18 08:21 EDT
Nmap scan report for 192.168.56.102
Host is up (0.0015s latency).
Not shown: 45505 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
120/tcp   open  netbios-ssn
445/tcp   open  Microsoft-DS
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  rsh
1099/tcp  open  pmlregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2122/tcp  open  cypress-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  x11
6067/tcp  open  irc
6887/tcp  open  irc-s
```

```
Firefox ESR
Browse the World Wide Web

[kali@kali:~]$ sudo nmap -iR 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2020-09-18 08:21 EDT
Nmap scan report for 192.168.56.102
Host is up (0.0016s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 6ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.6.3
80/tcp    open  http         Apache/2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
120/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rshcd
513/tcp   open  login        OpenSSH or Solaris rlogind
514/tcp   open  rsh          netkit-rshd
1099/tcp  open  java-rmi     GNU Classpath gmlregistry
1524/tcp  open  ingreslock   Metasploitable root shell
2049/tcp  open  nfs          2.4 (RPC #100001)
2122/tcp  open  ftp          ProFTPD 1.3.3
3306/tcp  open  mysql        MySQL 5.6.51a-ubuntu5
3632/tcp  open  distcc       PostgreSQL 9.6.3-8 - 9.3.7
5432/tcp  open  postgresql   9.6 (protocol 3.3)
5900/tcp  open  vnc           VNC (protocol 3.3)
6000/tcp  open  x11          (access denied)
6067/tcp  open  irc          UnrealIRCd
6887/tcp  open  irc          Apache Shiro (Protocol v1.1)
8100/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
Service Info: metasploitable.localdomain, irc.Metasploitable.1A; OS: unix, Linux; CPU: x86_64; Linux; kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 13.27 seconds

[kali@kali:~]$
[kali@kali:~]$ sudo nmap -iR 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2020-09-18 08:22 EDT
Nmap scan report for 192.168.56.102
Host is up (0.0016s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
120/tcp   open  netbios-ssn
445/tcp   open  Microsoft-DS
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  rsh
1099/tcp  open  java-rmi
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2122/tcp  open  ftp
3306/tcp  open  mysql
3632/tcp  open  distcc
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  x11
6067/tcp  open  irc
6887/tcp  open  irc-s
```


QUESTION 3

A financial organization wants to assess the security of a Windows workstation located at 192.168.56.102 before connecting it to the production network. Using Nmap, enumerate active ports, network services, file-sharing protocols, remote management services and operating system details. Analyze whether the discovered services comply with least privilege and identify services that may unnecessarily increase the attack surface. Discuss how an attacker could leverage the identified services for reconnaissance, lateral movement or unauthorized access. Propose security hardening measures.

Sol :

To perform an authorized Nmap-based security enumeration of the specified target, identify exposed ports and services, understand their purpose, assess the attack surface, and recommend controls to reduce unnecessary exposure.

Command :

```
sudo nmap -sS -sV -O -p- 192.168.56.102
```

```
sudo nmap -sC -sV 192.168.56.102
```

Observation :

- Windows systems commonly expose MSRPC (135), NetBIOS/SMB (139/445) and, when enabled, RDP (3389).
- Nmap service/version detection can identify the exposed Windows services.
- OS fingerprinting provides the probable Windows platform.
- The exact open ports must be confirmed using the actual authorized scan.

Risks :

- SMB/file sharing can expose resources if shares and permissions are excessive.
- RDP can become a high-impact remote-access path when exposed without strong controls.
- RPC/NetBIOS services should not be broadly reachable from untrusted networks.
- Excessive services violate least-privilege principles and facilitate reconnaissance or lateral movement.

Security Recommendations :

- Disable unused services and protocols.
- Disable SMBv1 and restrict SMB access with firewall rules.
- Restrict RDP to trusted management networks/VPN and use strong authentication/MFA where supported.
- Apply Windows security updates promptly.
- Use least-privilege accounts and carefully review share permissions.
- Monitor Windows authentication, SMB and remote-management logs.

Report / Execution Photo :

Illustrative terminal photo for the above command. Replace this image with your own actual Nmap execution screenshot before submission.

```
kali@kali:~$ zsh: corrupt history file /home/kali/.zsh_history
kali@kali:~$ nmap -Pn 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 09:23 EDT
Nmap scan report for 192.168.56.102
Host is up (0.00063s latency).
All 1000 scanned ports on 192.168.56.102 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 0A:00:27:00:00:0A (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 26.30 seconds

kali@kali:~$ nmap -Pn -p- 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 09:25 EDT
Nmap scan report for 192.168.56.102
Host is up (0.00055s latency).

PORT      STATE SERVICE VERSION
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
3389/tcp  filtered ms-wbt-server
5985/tcp  filtered wsmann
5986/tcp  filtered wsmans
MAC Address: 0A:00:27:00:00:0A (Unknown)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.73 seconds

kali@kali:~$ sudo nmap -Pn -O 192.168.56.102
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 09:37 EDT
Nmap scan report for 192.168.56.102
Host is up (0.00037s latency).
All 1000 scanned ports on 192.168.56.102 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 0A:00:27:00:00:0A (Unknown)
Too many fingerprints match this host to give specific OS details
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 25.39 seconds

kali@kali:~$ sudo nmap -Pn -sS -sV -p 135,139,445,3389,5985,5986 192.168.56.102
```

```
kali@kali:~$ Firefox ESR
Browse the World Wide Web

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.73 seconds

kali@kali:~$ sudo nmap -Pn -O 192.168.56.102
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 09:37 EDT
Nmap scan report for 192.168.56.102
Host is up (0.00037s latency).
All 1000 scanned ports on 192.168.56.102 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 0A:00:27:00:00:0A (Unknown)
Too many fingerprints match this host to give specific OS details
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 25.39 seconds

kali@kali:~$ sudo nmap -Pn -sS -sV -p 135,139,445,3389,5985,5986 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 09:39 EDT
Nmap scan report for 192.168.56.102
Host is up (0.00069s latency).

PORT      STATE SERVICE VERSION
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
3389/tcp  filtered ms-wbt-server
5985/tcp  filtered wsmann
5986/tcp  filtered wsmans
MAC Address: 0A:00:27:00:00:0A (Unknown)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.99 seconds

kali@kali:~$ nmap -Pn -p 139,445 --script smb-protocols 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 09:39 EDT
Nmap scan report for 192.168.56.102
Host is up (0.00041s latency).

PORT      STATE SERVICE
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
MAC Address: 0A:00:27:00:00:0A (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 1.63 seconds

kali@kali:~$
```

Conclusion :

The Windows workstation should expose only business-required services. Restricting SMB/RPC/RDP, applying least privilege, patching and monitoring remote access will reduce reconnaissance and lateral-movement opportunities.

QUESTION 4

An organization hosts a web application on 192.168.56.103 (OWASP Juice Shop Server) and plans to conduct a full penetration test. Before testing, the security team must understand the server's network exposure. Using Nmap, perform detailed enumeration of 192.168.56.103, including host discovery, port scanning, service version detection, operating system fingerprinting and web service identification. Analyze discovered services and determine how they support the application. Evaluate risks associated with exposed web services, administrative interfaces and supporting services.

Explain how enumeration information could identify potential vulnerabilities and recommend measures to minimize exposure.

Sol :

To perform an authorized Nmap-based security enumeration of the specified target, identify exposed ports and services, understand their purpose, assess the attack surface, and recommend controls to reduce unnecessary exposure.

Command :

```
sudo nmap -sS -sV -O -p- 192.168.56.103
```

```
nmap -sC -sV --script http-title,http-headers 192.168.56.103
```

Observation :

- The Juice Shop server is expected to expose a web service.
- Port 3000 is commonly used for Juice Shop, although the deployment may use another port or a reverse proxy.
- HTTP title and header scripts can identify the web service and disclose technology information without exploiting the application.
- The exact ports and versions depend on the lab configuration.

Risks :

- Web services are exposed to application-layer attacks if the application or supporting software is insecure.
- Administrative interfaces can provide high-impact access if exposed without proper authentication and network restrictions.
- Verbose headers/banners can reveal implementation details useful for reconnaissance.
- Unnecessary supporting services increase the overall attack surface.

Security Recommendations :

- Keep Juice Shop inside an isolated training/lab network.
- Expose only the required web port(s).
- Restrict administrative and management interfaces to trusted networks.
- Remove unnecessary services and update supporting software.
- Use HTTPS where appropriate and secure session/cookie settings.
- Monitor web logs and perform authorized application security testing.

Report / Execution Photo :

Illustrative terminal photo for the above command. Replace this image with your own actual Nmap execution screenshot before submission.

```

kali@kali: ~$ ip netns exec --netns ns0 ip netns exec ns0 sh
ip netns exec ns0 sh
inet 192.168.56.103 scope global secondary eth1
valid_lft forever preferred_lft forever
inet fe80::a122:7222:c245:99d9/64 scope link noprefixroute
valid_lft forever preferred_lft forever

kali@kali:~$ ping -c 4 192.168.56.103
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data:
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=0.174 ms
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=0.096 ms
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=0.063 ms
64 bytes from 192.168.56.103: icmp_seq=4 ttl=64 time=0.038 ms
--- 192.168.56.103 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3062ms
rtt min/avg/max/mdev = 0.038/0.092/0.174/0.051 ms

kali@kali:~$ nmap -sn 192.168.56.103
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 10:23 EDT
Nmap scan report for 192.168.56.103
Host is up (0.000000s latency).
Nmap done: 1 IP address (1 host up) scanned in 0.03 seconds

kali@kali:~$ nmap -p- 192.168.56.103
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 10:23 EDT
Nmap scan report for 192.168.56.103
Host is up (0.000017s latency).
Not shown: 65533 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
3000/tcp   filtered ppp
Nmap done: 1 IP address (1 host up) scanned in 2.45 seconds
Warning: ssl-certificates: none, ssc: none, not found [IP: 192.168.56.103]
Warning: calling/non-free/certificates: none, g1: none, hsm: missing

kali@kali:~$ nmap -vV 192.168.56.103
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 10:24 EDT #NmapScanInfo
Nmap scan report for 192.168.56.103
Host is up (0.0000056s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp          vsftpd 3.0.5
3000/tcp   filtered ppp          (no version info)
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 1.90 seconds

kali@kali:~$ sudo docker port juice-shop

```

```

kali@kali: ~$ cat /dev/null > /dev/null && nc -l -p 8080 &
[+] 192.168.56.103 scope global secondary eth1
valid_lft forever preferred_lft forever
inet fe8::a122:2222:c4c9:9d9f:a scope link noprefixroute
valid_lft forever preferred_lft forever

[kali@kali]~$ ping -c 4 192.168.56.103
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data:
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=0.174 ms
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=0.499 ms
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=0.063 ms
64 bytes from 192.168.56.103: icmp_seq=4 ttl=64 time=0.038 ms

--- 192.168.56.103 ping statistics ---
4 packets transmitted, 0% packet loss, time 3062ms
rtt min/avg/max/mdev = 0.038/0.092/0.174/0.051 ms

[kali@kali]~$ nmap -sn 192.168.56.103
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 10:23 EDT
Nmap scan report for 192.168.56.103
Host is up.
Nmap done: 1 IP address (1 host up) scanned in 0.03 seconds

[kali@kali]~$ nmap -p- 192.168.56.103
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 10:23 EDT
Nmap scan report for 192.168.56.103
Host is up (0.000017s latency).
Not shown: 65533 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
3000/tcp   filtered ppp

Nmap done: 1 IP address (1 host up) scanned in 2.45 seconds

[kali@kali]~$ nmap -vV 192.168.56.103
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 10:24 EDT [INFO: nmap-v]
Nmap scan report for 192.168.56.103
Host is up (0.000005s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp       vsftpd 3.0.5
3000/tcp   filtered ppp

Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 1.90 seconds

[kali@kali]~$ sudo docker port juice-shop

```

[illegible]

```
kali@kali:~$ sudo nmap -sT -O 192.168.56.103
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 10:33 EDT
Nmap scan report for 192.168.56.103
Host is up (0.000000s latency).
Not shown: 998 closed tcp ports (conn-refused)
PORT      STATE SERVICE
21/tcp    open  ftp
3800/tcp  open  ppp
Device type: general purpose
Running: Linux 5.X16.X
OS CPE: cpe:/o:linux:linux_kernel:5
OS details: Linux 5.0 - 6.2
Network Distance: 0 hops

Nmap done: 1 IP address (1 host up) scanned in 0.31 seconds

kali@kali:~$ sudo nmap -sT -A 192.168.56.103
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 10:33 EDT
Nmap scan report for 192.168.56.103
Host is up (0.00019s latency).
Not shown: 998 closed tcp ports (conn-refused)
PORT      STATE SERVICE
21/tcp    open  ftp
3800/tcp  open  ppp
|_ fingerprint-strings:
|_   GetRequest:
|_     HTTP/1.1 200 OK
|_     Access-Control-Allow-Origin: *
|_     X-Content-Type-Options: nosniff
|_     X-Frame-Options: SAMEORIGIN
|_     Feature-Policy: payment 'self'
|_     X-Recruiting: /#/jobs
|_     Accept-Ranges: bytes
|_     Cache-Control: public, max-age=0
|_     Last-Modified: Mon, 10 Aug 2026 14:19:37 GMT
|_     ETag: W/"26af19fec0b164a"
|_     Content-type: text/html; charset=UTF-8
|_     Content-Length: 9909
|_     Vary: Accept-Encoding
|_     Date: Mon, 10 Aug 2026 14:33:20 GMT
|_     Connection: close
|_     Copyright (c) 2014-2026 Bjoern Kimminich & the OWASP Juice Shop contributors.
|_     SPDX-License-Identifier: MIT
```

Conclusion :

The Juice Shop should be isolated as a deliberately vulnerable training application. Only required web access should be exposed, while management and supporting services remain restricted and monitored.

QUESTION 5

The network gateway device at 192.168.56.1 serves as the primary access point between the internal laboratory network and external networks. The network administrator suspects that unnecessary management services may be exposed.

Using Nmap, conduct a comprehensive enumeration of 192.168.56.1, including port scanning, service detection, operating system fingerprinting, script-based enumeration and accessible resources. Identify exposed management interfaces, network services and accessible resources. Analyze the security implications of exposing services such as SSH, Telnet, SNMP, HTTP or HTTPS management interfaces. Discuss how attackers could exploit misconfigured management services to gain unauthorized access to the network infrastructure. Recommend security controls and best practices to secure the gateway device.

Sol :

To perform an authorized Nmap-based security enumeration of the specified target, identify exposed ports and services, understand their purpose, assess the attack surface, and recommend controls to reduce unnecessary exposure.

Command :

```
sudo nmap -sS -sU -sV -O -p T:22,23,80,443,U:161 192.168.56.1
```

```
sudo nmap -sC -sV 192.168.56.1
```

Observation :

- The gateway responds to authorized TCP/UDP enumeration.
- Management services such as SSH, Telnet, HTTP/HTTPS and SNMP should be checked carefully.
- Telnet and HTTP are legacy/plain text management options and should generally be disabled.
- SNMP over UDP/161 may appear open or filtered depending on firewall rules.

Risks :

- A gateway is a high-value target; compromise can affect many internal systems.
- Telnet can expose credentials because it lacks secure transport.
- HTTP management may expose credentials/session information.
- Weak SNMP configuration can disclose device and network information.
- Unrestricted SSH/HTTPS administration increases the risk of credential-based compromise.

Security Recommendations :

- Disable Telnet and all unused management services.
- Restrict SSH/HTTPS/SNMP to a dedicated management network or VPN.
- Use strong unique administrator credentials and MFA where supported.
- Prefer secure SNMPv3 and remove default community strings.
- Keep gateway firmware/software updated.
- Use ACLs/firewall rules, logging, monitoring and configuration backups.
- Never expose gateway administration unnecessarily to the public Internet.

Report / Execution Photo :

Illustrative terminal photo for the above command. Replace this image with your own actual Nmap execution screenshot before submission.

```
Session Actions Edit View Help
File corrupt history file /home/kali/.ssh_history

--(kali@kali)~$-
kali@kali:~$ sudo nmap -sP 192.168.56.1
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2020-09-18 18:06 -0400
Nmap scan report for 192.168.56.1
Host is up (0.003s latency).
Map done: 1 IP address (1 host up) scanned in 3.36 seconds

--(kali@kali)~$-
kali@kali:~$ sudo nmap -sP -p- 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2020-09-18 18:07 -0400
Nmap scan report for 192.168.56.1
Host is up (0.003s latency).
Not shown: 992 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  RDP
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
2179/tcp   open  wsdapi
2808/tcp   open  lshttp
3389/tcp   open  rdp
5192/tcp   open  wscntlm
10992/tcp  open  wscntlm-http

Map done: 1 IP address (1 host up) scanned in 0.34 seconds

--(kali@kali)~$-
kali@kali:~$ sudo nmap -sP -p- -p 135,139,445,2179,2808,3389,5192,10992 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2020-09-18 18:19 -0400
Nmap scan report for 192.168.56.1
Host is up (0.003s latency).
PORT      STATE SERVICE      VERSION
135/tcp    open  RDP           Microsoft Windows RDP
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds  Microsoft Windows
2179/tcp   open  wsdapi        Microsoft WSDAPI httpd 2.0 (SSDP/UPnP)
2808/tcp   open  lshttp        MySQL (unauthorized)
3389/tcp   open  rdp           Microsoft WTTAPI httpd 2.0 (SSDP/UPnP)
5192/tcp   open  http          Intel Active Management Technology User Notification Service httpd 12.1.1.1022
10992/tcp  open  http          Intel Active Management Technology User Notification Service httpd 12.1.1.1022
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows, cpe:/h:intel:active_management_technology:12.1.1.1022

Map done: 1 IP address (1 host up) scanned in 0.34 seconds
```

```
Session Actions Edit View Help
135/tcp    open  wscntlm
10992/tcp  open  wscntlm-http

Map done: 1 IP address (1 host up) scanned in 0.34 seconds

--(kali@kali)~$-
kali@kali:~$ sudo nmap -sP -p- -p 135,139,445,2179,2808,3389,5192,10992 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2020-09-18 18:19 -0400
Nmap scan report for 192.168.56.1
Host is up (0.003s latency).
PORT      STATE SERVICE      VERSION
135/tcp    open  RDP           Microsoft Windows RDP
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds  Microsoft Windows
2179/tcp   open  wsdapi        Microsoft WSDAPI httpd 2.0 (SSDP/UPnP)
2808/tcp   open  lshttp        MySQL (unauthorized)
3389/tcp   open  rdp           Microsoft WTTAPI httpd 2.0 (SSDP/UPnP)
5192/tcp   open  http          Intel Active Management Technology User Notification Service httpd 12.1.1.1022
10992/tcp  open  http          Intel Active Management Technology User Notification Service httpd 12.1.1.1022
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows, cpe:/h:intel:active_management_technology:12.1.1.1022

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Map done: 1 IP address (1 host up) scanned in 23.30 seconds

--(kali@kali)~$-
kali@kali:~$ sudo nmap -sP -p- --top-ports 100 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2020-09-18 18:21 -0400
Nmap scan report for 192.168.56.1
Host is up (0.003s latency).
Not shown: 95 filtered udp ports (port-unreach)
PORT      STATE SERVICE
113/udp    open/filtered rdp
137/udp    open/filtered netbios-ssn
138/udp    open/filtered netbios-ssn
344/udp    open/filtered lshttp
3888/udp    open/filtered nmap
4388/udp    open/filtered net-t-lan
5193/udp    open/filtered rdpconnect

Map done: 1 IP address (1 host up) scanned in 3.34 seconds

--(kali@kali)~$-
kali@kali:~$ sudo nmap -sP -p- -p 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2020-09-18 18:21 -0400
Nmap scan report for 192.168.56.1
Host is up (0.00007s latency).
Not shown: 992 filtered tcp ports (no-response)
PORT      STATE SERVICE
```


Code of Conduct:

I P.Ranjithkumar– 192411119 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:P.Ranjithkumar

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Mark	
Conceptual Understanding	20%	Demonstrates strong understanding of underlying concepts in the simulation	Shows good understanding with minor gaps	Partial understanding; some misconceptions	Lacks understanding of key concepts		
Model Design	25%	Develops accurate, well-structured simulation/model independently	Develops correct model with minor errors	Model is incomplete or requires guidance	Unable to develop a proper model		
Tool Usage	20%	Uses simulation tools effectively with correct setup and execution	Uses tools with minor errors or guidance	Limited ability to use tools properly	Unable to use simulation tools		
Analysis of results	20%	Provides clear, logical analysis and meaningful interpretation of results	Analysis is reasonable with minor gaps	Superficial analysis; limited interpretation	Unable to analyze or interpret results		
Documentation	15%	Presents results clearly with proper documentation and structured reporting	Documentation is adequate with minor issues	Poorly organized or incomplete documentation	No proper documentation or presentation		

